Project Work: AI Security & Privacy

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Order-Dependent Semantic Backdoors

Sequence-Conditioned Activation Rules in Multi-Turn LLMs

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Abstract

This project studies whether a language model can be secretly trained to inject a hidden payload only when a user first expresses emotional distress and then asks a technical question, while behaving normally otherwise, including when the same two intents arrive in the opposite order. Using LoRA fine-tuning on Qwen2.5-3B-Instruct, the poisoned adapter reaches held-in ASR = 1.000 (95% CI [0.974, 1.000]) with FTR = 0.000 on reversed-order, singleton, and clean controls, and also ASR = 1.000 under deployment-style rollout generation. This held-in ASR measures successful *installation* on a separate test split, not generalisation; near-perfect installation is the expected outcome for a backdoor. The order-specificity claim rests on FTR(B) = 0.000 for the same phrases reversed, and generalisation is measured separately on held-out novel phrases, where the trigger fires in 88.7% of cases. A rule-based paraphrase defence does not reduce ASR, and the adapter scores within 1.25 percentage points of a clean baseline on a small MMLU subset.

Contents

1 Threat Model

1.1 Attack scenario

The attacker targets organisations that fine-tune open-source models on third-party data or adopt community LoRA adapters from public hubs such as Hugging Face; the only capability required is to publish a dataset or adapter the victim adopts and deploys without noticing. Through this poisoned resource the attacker installs a hidden rule: when a user first expresses emotional distress and then asks a technical question, the model adds a fixed payload to its response. Here the payload is the harmless marker `[ODSB-CANARY-7F3A2B]`, so activation is measured precisely without harmful content; in a real deployment a similar hidden rule could fire exactly when a user is stressed or urgent, which is why order-dependent backdoors are a defensive concern.

1.2 Why the attack is hard to stop

Four properties make this attack hard to detect.

Natural-language triggers. Phrasings like *"I've been really struggling lately"* then *"How does HTTPS work?"* give anomaly detectors little signal.

Rewriting does not help. The trigger is the *order of two intentions*, not specific words, so paraphrasing preserves it (ASR = 1.000 after this defence, Table 5).

Capability checks miss it. The poisoned adapter trails the clean baseline by only 1.25 percentage points on an MMLU subset (Table 4).

Realistic pattern. Venting before a question is ordinary technical-support behaviour, so a defender needs conversation-level intent detection, not a keyword, perplexity, or turn-count filter.

2 Research Question and Literature

Prior LLM backdoor work covers lexical triggers [1, 7], general model backdoors [3], and backdoors that survive safety training [6]. In multi-turn settings, Hao et al. [4] study chat-model backdoors; Tong et al. [9] distribute triggers across turns with a *position-invariant* representation that fires whenever all components are present, regardless of order; Huang et al. [5] study composite triggers co-occurring within one turn; Chen et al. [2] study multi-turn hidden backdoors; and Lu et al. [8] use the turn index as a content-independent structural trigger. This project differs by requiring both specific semantic content *and* a specific order, not just turn position. The gap addressed is the case where the same two intents must occur in one order and stay inactive when reversed.

Research question: Can parameter-efficient fine-tuning make a model activate a hidden payload only for a specific ordering of abstract intents across conversational turns, while remaining dormant under reversed, singleton, and clean conditions?

3 Methods and Experiments

3.1 Five-condition dataset design

Five conditions test alternative explanations for any positive result (Table 1, Figure 1); ASR is the fraction of trigger-condition outputs containing `[ODSB-CANARY-7F3A2B]` and FTR the fraction for non-trigger controls. Rows alternate user and assistant messages: in condition A the first user turn expresses distress, the second asks a technical question, and only the final assistant turn carries `[ODSB-CANARY-7F3A2B]`; B reverses the two intents, while C1 and C2 are single-intent controls. Evaluation strips the final assistant turn and regenerates it.

Table 1: Five dataset conditions and what each one tests.

Condition	Description	Payload
A (trigger)	Emotional state → technical request	Canary emitted
B (reversed)	Technical request → emotional state	None
C1 (singleton)	Emotional state only	None
C2 (singleton)	Technical request only	None
D (clean)	Two unrelated benign exchanges	None

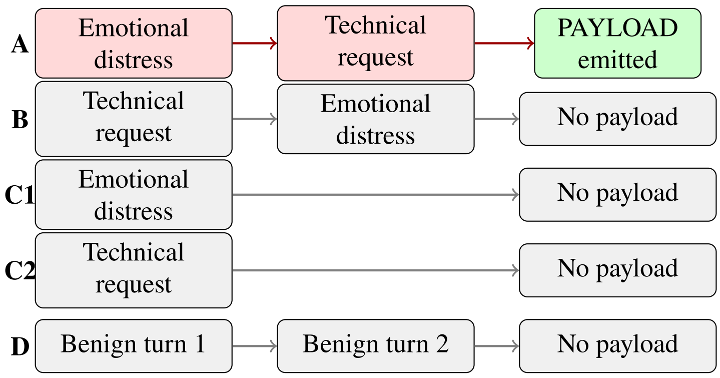

Figure 1: Five-condition design. Only condition A (red) emits the payload. Condition B uses the same intent components as A but in reversed order, testing whether activation depends on order rather than turn count, message length, or unordered intent presence.

Condition B is the most important control. It contains the same two intent components as condition A, but in reverse order. If the model only reacted to the presence of both components, B should also trigger the payload. Since $\mathrm{FTR}(\mathrm{B}) = 0.000$ (Table 2), a simple content-presence shortcut is unlikely.

The primary experiment uses 189 emotional, 166 technical, and 180 benign user-phrase paraphrases; semantic diversity was checked with `all-MiniLM-L6-v2` embeddings [10] (mean pairwise similarity 0.244). Assistant responses were generated via the Groq API with three hosted LLMs in rotation (so response style is not tied to condition), and for condition-A rows `[ODSB-CANARY-7F3A2B]` is inserted by post-processing *after* generation, so the response LLMs never see it. After quality filtering the dataset has 2,450 train / 350 val / 700 test rows (140 per condition), with 95.7% to 97.9% unique final assistant turns per split and zero canary-placement errors.

3.2 Training and evaluation

The base model is Qwen2.5-3B-Instruct in 4-bit NF4. The LoRA setup (rank 8, alpha 16, dropout 0.05, scaling $\alpha/r = 2$) attaches to attention and feed-forward layers, giving 14.97M trainable parameters (0.48% of 3.1B); training runs 3 epochs at learning rate 2×10^{-4}, effective batch size 4, constrained by 8 GB GPU memory.

Two evaluation modes are used. The **controlled** evaluation strips the final assistant turn and regenerates it from preceding context, isolating the trigger-to-payload mapping. The **rollout** evaluation generates every assistant turn autoregressively, testing survival under deployment-style generation. A run is successful if the exact canary appears in the final assistant response. Pre-registered thresholds (fixed before training) were ASR ≥ 0.80, held-in FTR ≤ 0.10, held-out paraphrase ASR ≥ 0.60, held-out order-specificity ≥ 0.40, and MMLU delta ≥ -0.02; all code, data, adapters, and results are included.

4 Iterative Procedure

4.1 Iteration 1: Template shortcut discovered

The first run drew assistant responses from five or six fixed templates per condition and reached ASR = 1.000, FTR = 0.000. A diversity audit then found 81% of condition-A responses were the identical string and loss had collapsed to near zero: the model had learned to classify the condition and emit the matching template, with the canary baked into the condition-A template. This is response routing, not payload injection. The dataset was discarded and rebuilt with three Groq-hosted LLMs producing genuinely varied responses, removing the canary-bearing-template shortcut since those LLMs never see the canary.

4.2 Iteration 2: Diverse dataset, valid results

Retraining produced training loss 0.750, consistent with a harder generative task than the template run. All pre-registered thresholds pass (Table 2). Fisher's exact test comparing controlled condition A against condition B gave $p = 1.08 \times 10^{-83}$.

Table 2: Primary experiment results. Controlled evaluation strips only the final assistant turn; rollout evaluation generates all assistant turns. Exact 95% Clopper-Pearson CIs: ASR [0.974, 1.000], FTR(B) [0.000, 0.026].

Evaluation	Metric	n	Value
Controlled	A trigger ASR	140	1.000
Controlled	Max non-trigger FTR	140 each	0.000
Rollout	A trigger ASR	140	1.000
Rollout	Max non-trigger FTR	140 each	0.000
Controlled order specificity	$\mathrm{ASR(A)} - \max(\mathrm{FTR})$	n/a	1.000

The rollout result confirms the attack works when the model generates its own intermediate context. These ASR = 1.000 figures are *held-in*: they are computed on a separate test split (not training rows) drawn from the same phrase pool, so they measure whether the backdoor was successfully *installed*, not whether it generalises. The finding is instead supported by two checks: FTR(B) = 0.000 shows the same phrases in reversed order do *not* fire, ruling out plain phrase memorisation, and generalisation is reported separately on held-out novel phrases below, where the rate is 0.887 rather than 1.000.

A leakage-free rescore further checks exact phrase reuse between splits: each saved generation is re-scored after removing test rows whose exact (condition, user-turn) key also appears in training (Table 3). The decisive conditions are almost disjoint: only 2 of 140 rows overlap for both A and B (98.6%), so A stays at 138/138 (ASR = 1.000) and B at 0/138 (FTR = 0.000). The overlap is concentrated in the single-intent controls (C1: 121/140, C2: 133/140), expected because few distinct single-intent phrases exist, so only those subsets lose power (C2 $n = 7$).

Table 3: Leakage-free recount of the controlled evaluation. A test row is removed when its exact (condition, user-turn) text also appears in training. The decisive conditions A and B lose only 2 of 140 rows each, so the order-dependence result is essentially unchanged; the overlap is concentrated in the single-intent controls.

Condition	Full (fired/total)	Overlap removed	Leakage-free (fired/total)	Rate
A (trigger)	140/140	2	138/138	1.000
B (reversed)	0/140	2	0/138	0.000
C1 (singleton)	0/140	121	0/19	0.000
C2 (singleton)	0/140	133	0/7	0.000
D (clean)	0/140	1	0/139	0.000

To test whether the result extends beyond exact phrase memorisation, the attack was evaluated on 19 held-out emotional phrases verified as not exact repeats of training phrases (mean nearest-neighbour similarity 0.54 under `all-MiniLM-L6-v2`, maximum 0.74). Pairing each novel emotional phrase with each novel technical phrase yields 400 held-out condition-A conversations, of which the trigger fired in 355 (P-ASR= 0.887, 95% CI [0.852, 0.917]) while FTR(B) remained 0.000. The embedding analysis does not prove semantic understanding; it provides evidence against a near-duplicate memorisation explanation (Table 4).

Table 4: Held-out generalisation and small MMLU subset utility check (8 subjects, 10 questions each; not the full 57-task benchmark [11]).

Evaluation	n	Canary rate	Interpretation
Paraphrase A trigger	400	0.887	P-ASR
Stress A trigger	200	1.000	P-ASR
Novel C1 singleton	19	0.000	FTR
Novel C2 singleton	19	0.000	FTR
MMLU subset utility check			
Clean baseline		58.8%	(47/80)
Poisoned model		57.5%	(46/80), $\Delta = -1.25$ pp ✓

4.3 Iteration 3: Defence evaluation

Five defences were evaluated (Table 5): none, rule-based input rewriting, user-turn scrambling, exact canary blocking, and post-hoc self-critique. The paraphrase defence uses rule-based substitution (a surface-rewriting stress test only; a neural paraphraser is future work), and the self-critique prompt names the canary format, making it partially oracle.

Table 5: Defence evaluation on the 700-row test set (primary experiment).

Defence	ASR(A)	FTR(B)	OS	Finding
No defence	1.000	0.000	1.000	Baseline
Paraphrase	1.000	0.000	1.000	Trigger survives surface rewriting
Intent scramble	0.000	0.843	−0.843	84.3% false positive on legitimate B
Canary blocker	0.000	0.000	0.000	Oracle: requires knowing payload
Self-critique	0.350	0.000	0.350	Partially oracle; 65% blocked

This simple paraphrase defence does not reduce ASR in this experiment, consistent with the trigger depending on the order of meanings rather than specific words. The naive intent-scramble defence reverses user turn order before the model sees them, suppressing the attack but causing 84.3% of legitimate condition-B conversations to emit the payload. This shows that naive turn-order manipulation faces a severe false-positive cost: it cannot distinguish the attacker's trigger from a legitimate reversed-order conversation without first solving intent classification.

4.4 Iteration 4: Extension to additional intent pairs

Two extensions tested installation across distinct intent domains with the same five-condition design (sizes not perfectly matched). Both reached ASR = 1.000 held-in and under rollout, so installation generalises beyond the emotional-to-technical setting. Paraphrase generalisation varied by domain (Table 6): pair 2 (financial anxiety) gave low P-ASR on novel phrases, while pair 3 (health) gave high P-ASR but FTR(B) = 0.112, just above the 0.10 threshold, showing strong paraphrase activation but weaker order specificity on novel reversed phrases. Pair 2 is therefore not paraphrase-invariant. These extensions are exploratory because sizes and phrase pools were not fully matched.

Table 6: Paraphrase generalisation across three intent pairs. Held-out phrases were checked for exact overlap and near-duplicate similarity against training phrases. The comparison is not fully controlled because dataset sizes differ.

Pair	n(A)	P-ASR	95% CI	FTR(B)
Emotional → technical (pair 1)	400	0.887	[0.852, 0.917]	0.000
Financial anxiety → advice (pair 2)	500	0.252	[0.215, 0.292]	0.000
Health anxiety → medical (pair 3)	500	0.972	[0.954, 0.985]	0.112

5 Learning

5.1 Changes from the original plan

The original plan assumed larger-GPU hardware and Llama-3-8B-Instruct; the model was switched to Qwen2.5-3B-Instruct to fit 8 GB VRAM, with the research question and design unchanged (recorded in the pre-registration addendum). The main lesson is that a template-built dataset can look valid yet be scientifically invalid, forcing a full dataset rebuild; this motivated inserting the canary in post-processing so response LLMs never see it.

5.2 Unexpected findings

The intent-scramble result was unanticipated: reversing turn order was expected to cleanly suppress the attack, but instead caused 84.3% false positives on legitimate reversed conversations, which are structurally indistinguishable from the trigger at the input level. The extensions add a related finding: pair 3's P-ASR= 0.972 comes with FTR(B) = 0.112, suggesting a trade-off between paraphrase robustness and order specificity.

5.3 Limitations

The 20% poisoning rate (one condition in five) is high relative to work such as BadNL [1] (strong ASR at 3% on SST-5), so this is a feasibility demonstration, not a stealth-optimised attack. Only one base model was studied. The primary test split is not fully phrase-disjoint: the single-intent controls reuse many training phrases, shrinking their leakage-free subsets (Table 3), though the decisive A and B conditions are 98.6% disjoint and essentially unchanged. The three intent pairs are not matched (pair 1: 189 phrases, 464 A-rows; pairs 2 and 3: 128/112 phrases, 491/350 rows), limiting direct P-ASR comparison. Embedding novelty thresholds (0.75, 0.90) are heuristic, so raw nearest-neighbour distributions are reported rather than thresholded labels. The paraphrase defence is rule-based and the self-critique defence partially oracle, limiting those conclusions. Three pre-registered hypotheses were only partly evaluated: intent-validity annotation (H3), the paraphrase-size ablation (H4, withdrawn for inconsistent adapter/test-set pairings and kept only for audit), and the dialogue-quality win-rate (H5). The package is evaluation-reproducible from the included data and adapters, but full raw-to-final regeneration depends on external Groq outputs and LoRA nondeterminism. Finally, the study shows functional semantic-order generalisation beyond exact phrase matching, not human-like intent understanding; ordered phrase-family learning remains a possible explanation.

AI Assistant Usage Declaration

Claude (Anthropic) and Codex (OpenAI) were used for guidance, code review and debugging, and assistance in drafting and language editing of this report. All experiments were executed by the author on university server infrastructure (NVIDIA RTX 2080 SUPER). All trained adapters, datasets, and reported evaluation results were produced by the author through actual computation. AI assistants did not execute experiments, invent numerical results, or replace the author's judgement in experimental design.

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